COEP TECHNOLOGICAL UNIVERSITY

(COEP TECH)

(Formerly College of Engineering Pune)

END Semester Examination

(CT-20014) Data Structures and Algorithms – II

Course: B.Tech , Semester IV

Branch: Computer Engineering

Academic Year: 2023-2024

Max.Marks:60

Duration: 3 Hours

Instructions:

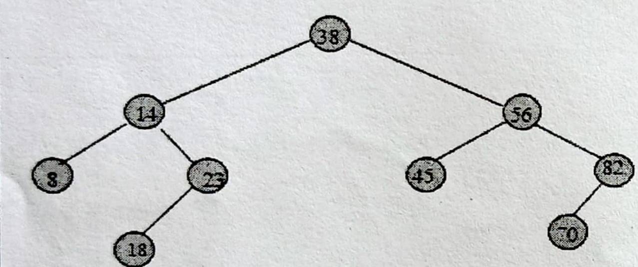
Student MIS No. 612203021

1. Figures to the right indicate the full marks.
2. Mobile phones and programmable calculators are strictly prohibited.
3. Writing anything on question paper is not allowed.
4. Exchange/Sharing of stationery, calculator etc. not allowed.
5. Write your MIS Number on Question Paper

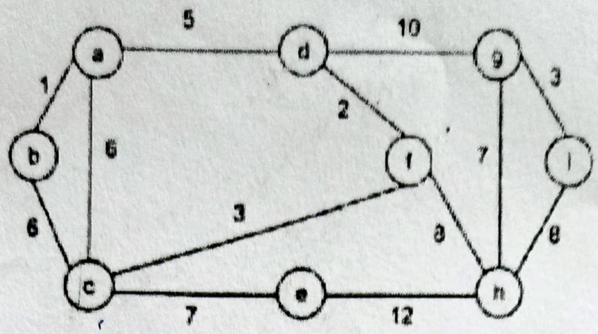
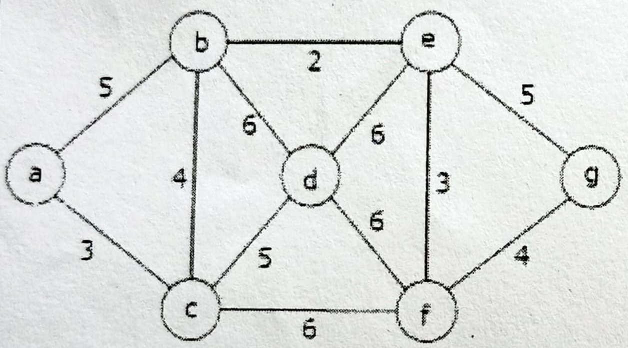
			Marks	CO	PO
Q 1	a	In a Binary Tree, how many max possible nodes can be at level=i?	1	1, 2	1, 2

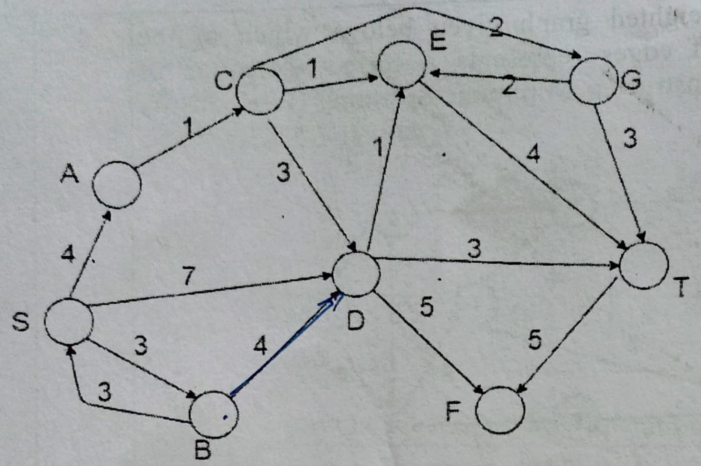
	b	Identify the following function and write its output
		`int function2(tree t)`
		`{`
		`stack a;`
		`init(&a);`
		`node *p=*t;`
		`int flag = 1;`
		`while(flag)`
		`{`
		`if(p)`
		`{`
		`push(&a,p);`
		`p = p->left;`
		`}`
		`else`
		`{`
		`if(!isEmpty(&a))`
		`{`
		`p = pop(&a);`
		`printf("%d ",p->data);`
		`p = p->right;`
		`}`
		`else`
		`flag = 0;`
		`}`
		`}`
		`}`
		The tree t is a binary search tree, having data inserted in the given order: 75, 68, 89, 21, 72, 80, 96. Make suitable assumptions wherever required.
			3	1, 2	1, 2

	c	Write a C function to display nodes of a binary tree level wise using queue. Assume linked implementation of the tree.	4	3, 4	1, 3
Q 2	a	Explain construction of Expression tree for given postfix expression: ab+cde+**	3	2, 3	1, 2
	b	Write an iterative function in C to delete node from a BST using linked list.	4	3, 4	1, 3

<table>
<tr><td>Q 3</td><td>a</td><td>Using Heaps, delete() can be implemented using in __________ time. Explain.</td><td>2</td><td>1, 2</td><td>1, 2</td></tr>
<tr><td></td><td>b</td><td>Consider the array A representing a min-heap below (only keys are shown) and answer the question that follows:
<table>
<tr><td>Array Index</td><td>1</td><td>2</td><td>3</td><td>4</td><td>5</td><td>6</td><td>7</td><td>8</td><td>9</td><td>10</td><td>11</td><td>12</td><td>13</td><td>14</td></tr>
<tr><td>Key</td><td>2</td><td>9</td><td>4</td><td>12</td><td>16</td><td>7</td><td>15</td><td>13</td><td>14</td><td>17</td><td>18</td><td>8</td><td>10</td><td>19</td></tr>
</table>
Give the array after performing one removeMin() operation.</td><td>2</td><td>1, 2</td><td>1, 2</td></tr>
<tr><td></td><td>c</td><td>Write a function in C to remove element in a heap using array with the process of heapify.</td><td>5</td><td>3, 4</td><td>1, 3</td></tr>
<tr><td>Q4</td><td>a</td><td>What is the maximum height of any AVL-tree with 7 nodes? Assume that the height of a tree with a single node is 0.</td><td>3</td><td>2, 3</td><td>1, 2</td></tr>
<tr><td></td><td>b</td><td>Consider the following AVL tree


What will be the resultant AVL tree after insertion of node with value 20?</td><td>3</td><td>2, 3</td><td>1, 2</td></tr>
<tr><td></td><td>c</td><td>Write a C function to insert node in AVL Tree using a linked list of nodes of structure having balance-factor, left pointer, right pointer, and parent pointers pointing to left sub-trees, right sub tree and parent of the node respectively.</td><td>5</td><td>3, 4</td><td>1, 3</td></tr>
<tr><td>Q5</td><td>a</td><td>In Red Black tree, operations such as insertion and deletion take __________ time and in skewed BST, take __________ time.</td><td>2</td><td>1, 2</td><td>1, 2</td></tr>
<tr><td></td><td>b</td><td>Show the Red-Black tree that results after each of the integer keys 12, 20, 6, 18, 19, 25 and 23 are inserted, in that order, into an initially empty Red-Black tree. Show each step.</td><td>3</td><td>2, 3</td><td>1, 3</td></tr>
<tr><td></td><td>c</td><td>i) What will be degree of a graph G with m edges.</td><td>1</td><td>1, 2</td><td>1, 2</td></tr>
<tr><td></td><td></td><td>ii) Given a weighted graph where weights of all edges are unique (no two edge have same weights), there is always a unique shortest path from a source to destination in such a graph. Explain.
(A) True
(B) False</td><td>2</td><td>1, 2</td><td>1, 2</td></tr>
</table>

Q6	a	For the undirected, weighted graph given below, which of the following sequences of edges represents a correct execution of Prim's algorithm to construct a Minimum Spanning Tree? Justify your answer  (A) - (a, b), (d, f), (f, c), (g, i), (d, a), (g, h), (c, e), (f, h) (B) - (c, e), (c, f), (f, d), (d, a), (a, b), (g, h), (h, f), (g, i) (C) - (d, f), (f, c), (d, a), (a, b), (c, e), (f, h), (g, h), (g, i) (D) - (h, g), (g, i), (h, f), (f, c), (f, d), (d, a), (a, b), (c, e)	4	1, 3	1, 2
	b	Consider the following graph:  Which one of the following is not the sequence of edges added to the minimum spanning tree using Kruskal's algorithm? Justify your answer. (A) (b,e)(e,f)(a,c)(b,c)(f,g)(c,d) (B) (b,e)(e,f)(a,c)(f,g)(b,c)(c,d) (C) (b,e)(a,c)(e,f)(b,c)(f,g)(c,d) (D) (b,e)(e,f)(b,c)(a,c)(f,g)(c,d)	4	1, 3	1, 2
	c	Consider the directed graph shown in the figure below. There are multiple shortest paths between vertices S and T. Which one will be reported by Dijstra's shortest path algorithm? Assume that, in any iteration, the shortest path to a vertex v is updated only when a strictly shorter path to v is discovered. Justify your answer.	4	2, 3	1, 2

<table>
<tr><td></td><td></td><td>

(A) SDT → 10
(B) SBDT → 10
(C) SACDT → 11.
(D) SACET → 10/11.</td><td></td><td></td><td></td></tr>
<tr><td></td><td>d</td><td>Write a function in C to initialize graph, check whether graph is connected and perform DFS from a randomly selected vertex.</td><td>5</td><td>3,4</td><td>1, 3</td></tr>
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